

PROPOSAL FOR ADOPTION

JCF DOCUMENT HUB

Mobile Access to Force Policies, Force Orders and Standard Operating Procedures

Prepared for
Jamaica Constabulary Force
Information & Communications Technology Division (ICTD)

Document Control

Document Title	JCF Document Hub — Proposal for Adoption (Version 2.0)
Version	2.0
Date Issued	13 May 2026
Classification	Confidential — For JCF Internal Review
Status	For Approval
Supersedes	Version 1.0 (22 April 2026)
Contributors	Sgt. Dwayne Aitken, Shevon Robinson, Nicholas Hunter
Email Address	dwayne.aitken@jcf.gov.jm, shevon.robinson@jcf.gov.jm, nicholas.hunter@jcf.gov.jm

Table of Contents

1. Executive Summary
2. Background and Problem Statement
3. Proposed Solution
4. Authentication via JCF Webmail
- 5. Access Control Model**
- 6. Backend API Specification**
7. Benefits to the Force
8. Security and Risk Considerations
- 9. Secure Local Storage Design**
- 10. Offline and Caching Strategy**
11. User Interface and Experience
- 12. Performance and Scalability**
- 13. Administration, Monitoring and DevOps**
- 14. Legal and Licensing**
15. Proposed Rollout Plan
16. Request to JCF
17. Closing Statement

Appendix A — Document Categories

Appendix B — AD Group to Document Access Matrix

Appendix C — API Endpoint Reference

Appendix D — Third-Party Library Licences

Appendix E — Privacy Notice and Data Minimisation Statement

Appendix F — Glossary

Appendix G — Application Screenshots

Note: Sections and appendices shown in green are new or substantially revised in Version 2.0.

1. Executive Summary

The JCF Document Hub is a native Android application that places the Force's policies, Force Orders, and Standard Operating Procedures directly in the hands of every serving officer — on duty, off duty, or at home. Officers authenticate with their existing @jcf.gov.jm webmail credentials and gain immediate, searchable access to the entire policy library on their personal smartphone.

The application has been designed, built, and tested. It currently indexes 346 documents across 19 categories, including Force Orders, divisional SOPs, and national policing policies. This proposal recommends a controlled pilot with a single division, followed by Force-wide rollout under ICTD sponsorship.

Version 2.0 of this proposal expands the original submission to address enterprise security, access control, API design, secure local storage, offline conflict handling, scalability, DevOps, and legal considerations raised during ICTD technical review. The following table summarises the additions.

Area	What was added in v2.0
Access control	End-to-end AD group to document mapping; server-side enforcement on every API call
Backend API	Full REST endpoint specification, PDF byte-range streaming, audit logging schema
Secure local storage	Android Keystore for tokens; encrypted app-private storage for cached PDFs; Intune remote wipe integration
Offline & caching	TTL policy, staleness indicators, conflict resolution on update
Performance & scalability	Pagination, CDN strategy, server-side caching, load testing plan
DevOps & monitoring	CI/CD pipeline, crash reporting, API versioning, configuration management
Legal & licensing	Third-party library licences, privacy notice, data minimisation statement

2. Background and Problem Statement

Officers across the Jamaica Constabulary Force routinely require access to policy documents to perform their duties correctly and lawfully. Examples include:

- A patrol officer responding to a missing-person report consulting the Missing Person Investigation SOP.
- A station commander preparing a disciplinary review consulting the Administrative Review Policy and SOP.
- A traffic officer requiring the latest Traffic and Mobile Maintenance Division (TMMD) operating procedures.
- A constable preparing for a promotional examination by studying current Force Orders.

In the current operating environment, accessing these documents typically requires:

- Physical presence at a station with access to the JCF network.
- Requesting copies via email from a supervisor or colleague.
- Reliance on printed binders that risk falling out of date.
- Searching through informal channels (e.g. WhatsApp threads) for forwarded copies.

This friction costs officer time, increases the risk of acting on outdated guidance, and prevents officers from preparing or studying outside of duty hours. It also poses a security risk: documents circulated informally via messaging platforms are outside the Force's control and cannot be versioned, recalled, or access-restricted.

3. Proposed Solution

The JCF Document Hub is a native Android application providing a single, authoritative point of access to all Force documentation.

Feature	Description
Centralised library	All 346 policy documents in one place, organised into 19 categories including Force Orders, NPCJ, TMMD, CIB, PMMD, SOPs, PECC, PRDB, PMAS, FLPD, FIPT, DWTT, FIBUA, PPMU, ICTD and ICT.
Role-based access	Officers see only the document categories their AD security group is authorised to access. Access is enforced server-side on every request.
Search	Full-text search by title across the entire library or within a specific category. Server-side index returns results only from documents the caller is authorised to view.
Integrated PDF viewer	Documents open directly within the application — pinch-to-zoom, page navigation, text selection, and bookmarks. No third-party reader required.
Read-aloud (Text-to-Speech)	Officers may have policy documents read aloud — beneficial during commutes, breaks, or for accessibility purposes.
Offline cache	Once a document has been opened, an encrypted copy is stored in app-private storage. It remains accessible without an active data connection. Cached copies carry a 7-day TTL.
Always current	Documents are served from a single ICTD-controlled source. When a policy is updated centrally, every device reflects the new version on next refresh or when the cached TTL expires.
Secure local storage	Authentication tokens are stored exclusively in the Android Keystore. Cached PDF files are stored in encrypted app-private storage, inaccessible to other applications.

4. Authentication via JCF Webmail

The application is gated behind the officer's existing JCF webmail credentials (@jcf.gov.jm). This approach delivers significant operational and security advantages.

4.1 Authentication Flow

Authentication follows the OAuth 2.0 Authorisation Code flow with PKCE, implemented via the Microsoft Authentication Library (MSAL) for Android:

1. The officer taps Sign In. The app generates a PKCE code verifier and challenge.
2. MSAL redirects to the Microsoft 365 login endpoint for the JCF tenant.
3. The officer enters their @jcf.gov.jm credentials. Microsoft 365 validates them against Azure Active Directory. MFA is enforced by Conditional Access policy.
4. On successful authentication, Microsoft issues an authorisation code to the app's registered redirect URI.
5. The app exchanges the code (plus PKCE verifier) for an access token and refresh token over HTTPS.
6. All subsequent API calls carry the access token in the Authorization: Bearer header. The backend validates the token against the JCF Microsoft 365 tenant on every request.

4.2 Security Properties

Property	Implementation
No new password to manage	Officers use credentials they already maintain for @jcf.gov.jm.
No parallel identity system	The JCF Microsoft 365 directory is the single source of truth.
Immediate deprovisioning	Disabling the webmail account automatically revokes application access. No separate offboarding step is required.
Domain restriction	Only @jcf.gov.jm addresses may authenticate. All others are rejected at the Microsoft 365 tenant level.
MFA enforcement	Microsoft 365 Conditional Access policies enforce multi-factor authentication, device compliance, and geo-restrictions automatically.
Token storage	Tokens are stored exclusively in the Android Keystore — never in SharedPreferences, local databases, or application files. They are not accessible to other applications or extractable without device unlock.
Credential transmission	Credentials are entered directly into Microsoft's hosted login page over TLS 1.2+. The application itself never sees or stores the password.
Certificate requirements	The app enforces TLS 1.2 minimum. Certificate pinning for the JCF API backend is applied in production to prevent MITM attacks on personal devices that may carry untrusted root CAs.

4.3 Session Management

Parameter	Value / Behaviour
Access token TTL	1 hour (set by Microsoft 365 tenant policy).
Refresh token TTL	90 days (set by Microsoft 365 tenant policy). Sliding window — resets on each use.
Silent refresh	MSAL silently refreshes the access token in the background before expiry. No officer action required during normal use.
Idle timeout	After 30 minutes of inactivity, the app requires the officer to re-authenticate. The offline cache remains accessible; no data is purged on idle timeout alone.
Explicit logout	Clears all tokens from Keystore, clears the MSAL cache, and deletes all cached PDF files from app-private storage.
Remote wipe	Microsoft Intune can remotely wipe the application or the entire device. This clears all tokens and cached documents.

5. Access Control Model

This section is new in Version 2.0. It defines how Active Directory security groups map to document categories, how that mapping is enforced on the backend, and how the model is administered.

5.1 Design Principles

- Access is deny-by-default. An officer who belongs to no relevant AD group sees no documents.
- All access control is enforced server-side on every API request. The client application is never trusted to self-report the caller's permissions.
- Group membership is resolved from the Azure AD token claims at request time. The backend does not cache group membership beyond a single request cycle unless a short-lived (5-minute) server-side cache is configured for performance.
- A nominated Content Owner (see Section 16) maintains the AD group assignments in consultation with the relevant division heads.

5.2 AD Group Structure

A dedicated Entra ID (Azure AD) security group is created for each document category, following the naming convention JCF-DocHub-{CATEGORY}. Officers are added to the appropriate groups by the ICTD administrator based on their role and division.

AD Security Group	Document Category	Eligible Officers
JCF-DocHub-General	General (statutes, acts)	All serving officers
JCF-DocHub-JCF	Force Orders, JCF Policies	All serving officers
JCF-DocHub-NPCJ	National Police College of Jamaica	Trainees and training staff
JCF-DocHub-TMMD	Traffic & Mobile Maintenance Division	TMMD personnel and supervisors
JCF-DocHub-PMMD	Public Morality & Major Decency Division	PMMD personnel and supervisors
JCF-DocHub-CIB	Criminal Investigation Branch	CIB detectives and supervisors
JCF-DocHub-PECC	Police Emergency Communications Centre	PECC operators and supervisors
JCF-DocHub-PRDB	Police Records & Data Branch	PRDB personnel and supervisors
JCF-DocHub-PMAS	Property Management & Admin Services	PMAS personnel and supervisors
JCF-DocHub-FLPD	Firearm Licensing & Permits Department	FLPD personnel and supervisors
JCF-DocHub-FIPT	Firearm Investigation &	FIPT investigators and supervisors

AD Security Group	Document Category	Eligible Officers
	Prosecution Team	
JCF-DocHub-DWTT	Detective & Witness Testimony Training	CIB detectives and training staff
JCF-DocHub-FIBUA	Fighting in Built-Up Areas	Tactical unit personnel — SSP authorisation required
JCF-DocHub-PPMU	Policy & Procedure Management Unit	PPMU staff and ICTD administrators
JCF-DocHub-ICTD	ICTD Procedures	ICTD staff and system administrators
JCF-DocHub-ICT	General ICT Guidelines	All serving officers
JCF-DocHub-SOPs	Cross-divisional SOPs	All serving officers
JCF-DocHub-SIMU	Strategic Intelligence Management Unit	SIMU personnel — ACP authorisation required
JCF-DocHub-Admin	All categories	ICTD administrators and Content Owner

5.3 Server-Side Enforcement

On every API request the backend performs the following steps before returning any document data:

7. Extract the Bearer token from the Authorization header.
8. Validate the token signature against the JCF Microsoft 365 JWKS endpoint.
9. Confirm the token has not expired and the audience claim matches the registered application ID.
10. Extract the groups claim from the validated token. If the claim is missing or empty, return HTTP 403 Forbidden.
11. For the requested document or category, look up the required group(s) from the access control configuration.
12. If the caller's groups claim does not include a required group, return HTTP 403 Forbidden. No document metadata or content is returned.
13. If authorised, stream the requested document or return the metadata list.

The client application is never trusted to communicate its own permission level. Steps 1–6 execute on every request, including requests for individual PDF pages.

6. Backend API Specification

This section is new in Version 2.0. It defines the RESTful API surface the mobile application uses to communicate with the JCF backend. All endpoints require a valid Bearer token and enforce the access control rules described in Section 5.

6.1 Base URL and Versioning

All endpoints are prefixed with `/api/v1/`. When a breaking change is required, a new version prefix (e.g. `/api/v2/`) is introduced alongside the existing one, giving clients a documented migration window of at least 90 days before the old version is retired.

6.2 Authentication Header

Every request must carry: `Authorization: Bearer {access_token}`

Requests without a valid token receive HTTP 401 Unauthorised. Requests with a valid token but insufficient group membership receive HTTP 403 Forbidden.

6.3 Endpoint Reference

Method + Path	Description	Notes
GET <code>/api/v1/categories</code>	List categories the caller is authorised to access.	Returns only categories where the caller's AD groups match. Response is paginated (default page size: 50).
GET <code>/api/v1/categories/{id}/documents</code>	List documents in a category.	Paginated. Supports query params: <code>page</code> , <code>page_size</code> (max 100), <code>sort</code> (name date), <code>order</code> (asc desc).
GET <code>/api/v1/documents/search?q={term}</code>	Full-text search across authorised documents.	Server-side index filters results to the caller's authorised categories. Returns document ID, title, category, and excerpt. Never returns content from unauthorised documents.
GET <code>/api/v1/documents/{id}/metadata</code>	Return metadata for a single document.	Includes: title, category, file size (bytes), last modified date, SHA-256 hash, version number. Returns 403 if caller lacks access.
GET <code>/api/v1/documents/{id}/content</code>	Stream the PDF document.	Supports HTTP Range header for byte-range requests. Enables progressive loading and resumable downloads. Returns <code>Content-Type: application/pdf</code> and <code>Content-Disposition: inline</code> .
GET <code>/api/v1/documents/{id}/thumbnail</code>	Return a JPEG thumbnail of the first page.	Generated server-side and cached for 24 hours. Max size: 256x256 px.
GET <code>/api/v1/sync/manifest</code>	Return a manifest of all authorised documents with hash and version.	Used by the app to detect documents that have changed since the last sync and trigger cache invalidation.

Method + Path	Description	Notes
POST /api/v1/audit/access	Record a document access event.	Called by the app when a document is opened or downloaded. Payload: {document_id, event_type, timestamp}. Server enriches with caller identity from token.

6.4 Rate Limiting

To prevent abuse and protect the document store, the following rate limits are enforced per authenticated user:

Limit type	Value
Document list requests	120 requests per minute
Document content (PDF stream)	30 requests per minute
Search requests	30 requests per minute
Thumbnail requests	60 requests per minute
Burst allowance	20 requests above limit before throttling begins
Throttle response	HTTP 429 Too Many Requests with Retry-After header
Bulk download detection	More than 50 distinct document downloads within 10 minutes triggers an alert to ICTD. The session is not automatically terminated, but the event is flagged for review.

6.5 Audit Logging

Every document access and download is written to an append-only audit log. The log schema is:

Field	Description
timestamp	UTC timestamp of the event (ISO 8601).
officer_email	The @jcf.gov.jm address from the validated token.
document_id	Internal document identifier.
document_title	Human-readable title at time of access.
event_type	One of: VIEW, DOWNLOAD, SEARCH, LOGIN, LOGOUT, FAILED_AUTH, ACCESS_DENIED.
client_ip	IP address of the request (for anomaly detection).
app_version	Version string of the mobile application.
outcome	SUCCESS or DENIED.

Audit logs are retained for a minimum of 2 years, stored in ICTD-controlled storage, and are not accessible from the mobile application. Log access requires separate administrator credentials.

7. Benefits to the Force

7.1 For the Rank-and-File Officer

- Study Force Orders at home in preparation for promotional examinations.
- Confirm the correct procedure on scene without leaving the location.
- Access the policy library on personal time, on a personal device, securely.
- Have documents read aloud during commutes (Text-to-Speech feature).

7.2 For Supervisors and Station Commanders

- Confidence that every officer holds the current, access-controlled version of every relevant policy.
- Elimination of printing, distribution, and acknowledgement chasing for paper-based policies.
- Simplified onboarding for new officers — the entire authorised policy library is one download away.

7.3 For the Force as an Institution

- Measurable reduction in policy-knowledge gaps among serving officers.
- A modern, professional tool that signals investment in officer development and welfare.
- Centralised version control — when a policy changes, every device in the Force updates within 7 days (cache TTL) or immediately on next connection.
- Full audit trail of document access for compliance and accountability purposes.

7.4 For ICTD

- Built on standard, supported technology (Flutter, OAuth 2.0, Microsoft 365, REST).
- No new servers required for the document library — content can be hosted on existing JCF-controlled SharePoint.
- Low operational overhead and a clear technology roadmap.
- Monitoring dashboards and crash reporting from day one.

8. Security and Risk Considerations

Version 2.0 expands this section significantly to address server-side enforcement, secure coding requirements, and data-at-rest protections.

Risk / Concern	Likelihood	Mitigation
Unauthorised access	Medium	JCF webmail (Microsoft 365) sign-in restricted to @jcf.gov.jm. MFA enforced via Conditional Access. Access denied to officers not in the relevant AD group.
Stale access for separated officers	High without control	Disabling the JCF webmail account immediately revokes token validity. The backend rejects any token whose account is disabled. No separate offboarding action is required.
Document leakage via sharing	Medium	Phase 2 controls: screenshot blocking, share restriction, and per-officer watermarking. In-transit, all content is served over TLS 1.2+.
Document integrity	Low	Documents served from a single ICTD-controlled source. Officers cannot modify or replace documents within the application. Each document exposes a SHA-256 hash for integrity verification.
Lost or stolen device	Medium	Re-authentication required after 30-minute idle period. Microsoft 365 Conditional Access (MFA, device compliance, remote wipe via Intune) applies automatically. Cached PDFs are stored encrypted; they cannot be read by other apps or extracted without device decryption.
MITM attack on personal devices	Low	Certificate pinning applied on the JCF API backend. The app rejects any TLS certificate not matching the pinned fingerprint.
Path traversal / injection attacks	Low	All document IDs are server-issued opaque identifiers (UUIDs). The API rejects requests that contain path separators or encoding attempts. User-supplied search queries are parameterised and never concatenated into file system paths.
Client-side bypass of access control	Low	The client application is never trusted to report its own permissions. The backend enforces group membership on every request independently.
Token theft from device	Low	Tokens are stored exclusively in the Android Keystore, which is hardware-backed on modern devices. Extraction without the device PIN/biometric is not feasible on non-rooted devices.
Bulk download / exfiltration	Low	Rate limiting (Section 6.4) and bulk download alerting (50+ documents in 10 minutes triggers ICTD alert) are in place. Audit logs provide forensic trail.

9. Secure Local Storage Design

This section is new in Version 2.0. It defines how the application stores sensitive data on the officer's personal device.

9.1 Authentication Tokens

Item	Storage approach
Access token	Stored by MSAL in Android Keystore-backed encrypted storage. Not accessible to other applications. Cleared on logout.
Refresh token	Stored by MSAL in Android Keystore-backed encrypted storage. Cleared on logout.
Officer identity (email)	Stored by MSAL alongside the token cache. Cleared on logout.
No credentials ever stored	The application never stores the officer's password. Credentials are entered via Microsoft's hosted login page.

9.2 Cached PDF Files

Item	Approach
Storage location	App-private internal storage (/data/data/{package}/files/cache/). Not accessible to other applications. Not accessible via Android file manager without root.
Encryption	Files are encrypted using AES-256-GCM. The encryption key is derived from a master key stored in the Android Keystore.
TTL (cache expiry)	7 days from the date of download. Expired files are deleted automatically on next app launch.
Version invalidation	When the sync manifest (Section 6.3) indicates a document has been updated server-side, the locally cached copy is deleted and re-fetched on next access.
Logout purge	All cached PDF files are deleted on explicit logout or when the Microsoft Intune remote wipe signal is received.
Maximum cache size	500 MB. When this limit is reached, the least-recently-used files are evicted first.

9.3 Device Wipe Integration (Intune)

Where the JCF deploys Intune mobile device management, the application registers with the Intune MAM (Mobile Application Management) SDK. This enables:

- Selective app wipe: ICTD can wipe only the Document Hub data (tokens + cached files) from a device without wiping the officer's personal data.
- Full device wipe: available when the device is enrolled in Intune MDM.

- Conditional Access compliance: devices that fail compliance checks (e.g. rooted device, out-of-date OS) are blocked from accessing documents, even with a valid token.

10. Offline and Caching Strategy

This section is new in Version 2.0. It defines how the application behaves when no data connection is available and how conflicts between cached and server-side content are resolved.

10.1 What is Cached

Data type	Caching behaviour
Category list	Cached in-memory on app launch. Refreshed from server on each new session. Not persisted to disk.
Document metadata list	Cached in the local database (Room, encrypted) for 24 hours per category. Displayed to the officer when offline.
PDF files	Cached to encrypted app-private storage on first open. 7-day TTL. Maximum 500 MB total.
Sync manifest	Fetches in the background every 6 hours when the device has connectivity. Used to detect updated documents without requiring a full re-download.

10.2 Offline Mode Behaviour

- When no connection is available, the app displays an Offline banner in the header.
- The officer can browse cached category lists and document metadata.
- Documents previously opened are available in full from the encrypted local cache.
- Documents not previously opened display a Not available offline notice.
- Search operates against the locally cached metadata only; full-text search across PDF content is not available offline.

10.3 Conflict Resolution

When the app re-connects after a period offline, the following process executes:

14. The app fetches the sync manifest from `/api/v1/sync/manifest`.
15. For each cached document, the local SHA-256 hash is compared to the hash in the manifest.
16. Documents where the hash differs are marked as Updated on server. The cached copy is immediately deleted.
17. When the officer next opens the updated document, the current version is fetched from the server.
18. A notification is shown: "X document(s) have been updated. Tap to refresh."

There is no merge or diff operation — policy documents are always authoritative from the server. The officer's locally cached copy is replaced in full.

10.4 Staleness Indicators

The application communicates cache status clearly in the UI:

State	UI indicator
Live (fetched this session)	No indicator — the default state.
Cached (from local storage)	Small "Cached" badge on the document tile. Shows date cached.
Expiring soon (within 24 hours)	"Expiring" badge shown in amber.
Expired (TTL elapsed without reconnect)	Document shown as Not available offline. Cannot be opened.
Updated on server	"Updated" badge shown in red. Cached copy deleted. Re-fetch required.

11. User Interface and Experience

This section expands the original application description to address the full file-explorer-like navigation model, accessibility requirements, and responsive design.

11.1 Navigation Model

Element	Description
Home screen	Summary tile (total documents, categories) and category grid. Tapping a category opens the document list for that category.
Breadcrumb bar	Displays the current path (e.g. Home > CIB > Documents). Back arrow and up-navigation are always visible.
Document list	Searchable, alphabetised list. Supports list view and grid view toggle. Documents show title, file size, last modified date, and cache status badge.
Context actions	Long-press or three-dot menu on any document reveals: Open, Download to cache, Properties (metadata), Share (Phase 2 — restricted). Delete from cache is available for locally cached files.
Downloaded files section	Dedicated section in the navigation drawer showing all locally cached documents grouped by category, with cache age and expiry.

11.2 Search

- Global search bar is always accessible from the toolbar.
- Server-side search queries the document index and returns results filtered to the officer's authorised categories. Results are never returned for unauthorised documents.
- Client-side search filters the currently loaded document list in real time as the officer types.
- Search results display: document title, category, file size, and a matched-text excerpt (server-side search only).

11.3 Accessibility

- All interactive elements have content descriptions for TalkBack (Android screen reader) support.
- Text-to-Speech reads entire PDF documents aloud at the officer's chosen speed.
- Minimum tap target size: 48 x 48 dp, consistent with Android accessibility guidelines.
- The application supports Android system font size scaling.
- Dark mode is supported — the application respects the system-level dark/light preference.

11.4 Responsive Design

The application is designed for portrait-mode phone use as the primary target. On tablets (screen width > 600 dp), the layout switches to a two-pane mode: category list on the left, document list on the right. The PDF viewer always occupies the full screen.

12. Performance and Scalability

This section is new in Version 2.0.

12.1 Current Scale

Metric	Current value
Documents indexed	346
Categories	19
Target concurrent users (pilot)	50–200 (one division)
Target concurrent users (Force-wide)	Up to 500 at peak shift-change times
Average PDF size	Estimated 500 KB–5 MB per document

12.2 Backend Performance Measures

Measure	Approach
Document list pagination	All list endpoints are paginated (default 50 items, max 100). The app uses infinite scroll — additional pages are fetched as the officer scrolls.
Server-side AD group caching	Azure AD group lookups are cached server-side with a 5-minute TTL per officer identity. This avoids redundant graph API calls on each request.
PDF byte-range streaming	PDFs are streamed progressively using HTTP Range requests. The PDF viewer renders visible pages immediately without waiting for the full file to download.
Thumbnail caching	First-page thumbnails are generated once and cached server-side for 24 hours. Regenerated only when the document is updated.
CDN / edge caching	Static document content (PDFs and thumbnails) is served via Azure CDN. Dynamic API responses (lists, metadata, search) are served directly from the application server. Cache-Control headers are set appropriately.
SharePoint throttling	If SharePoint is the document store, the backend implements retry logic with exponential back-off to handle Microsoft's per-tenant throttling limits.

12.3 Load Testing

Prior to Force-wide rollout (Phase 3), ICTD will conduct a load test simulating 500 concurrent users performing typical usage patterns (document list, search, PDF open). Acceptance criteria:

- P95 response time for document list: under 500 ms.
- P95 time-to-first-byte for PDF stream: under 1 second.
- Zero HTTP 5xx errors under normal load.
- Graceful degradation (HTTP 503 + Retry-After) under extreme load, with no data corruption.

12.4 Scalability Roadmap

The backend is containerised and deployable on Azure Container Apps or equivalent. Horizontal scaling (adding backend instances) is the primary scaling strategy. The stateless API design (all session state is held in the token) means any instance can serve any request.

13. Administration, Monitoring and DevOps

This section is new in Version 2.0.

13.1 Configuration Management

Configuration item	Storage and management
Azure AD tenant ID and application (client) ID	Environment variable, injected at build time via the CI/CD pipeline. Not hardcoded.
JCF API base URL (dev / staging / prod)	Environment variable, injected at build time.
Certificate pin fingerprint	Compiled into the release build. Requires an app update to rotate.
Feature flags (Phase 2 controls)	Controlled via a remote configuration service (e.g. Azure App Configuration). Enables screenshot blocking and watermarking to be activated without an app update.
Document access control matrix	Stored in version-controlled configuration on the backend. Changes require ICTD administrator approval and are applied without an app update.

13.2 Monitoring and Alerting

Metric / Event	Tooling
API error rates and latency	Azure Application Insights (or equivalent). Dashboard visible to ICTD. Alert triggers if P95 latency exceeds 2 seconds or error rate exceeds 1%.
Mobile app crashes	Firebase Crashlytics (free tier). Crash reports include stack trace, device model, OS version, and app version. No personally identifiable data is sent.
Bulk download alerts	Custom alert fires when any officer downloads more than 50 distinct documents within 10 minutes (see Section 6.4). Alert sent to ICTD duty administrator by email.
Failed authentication attempts	Surfaced via Microsoft 365 Sign-in logs. ICTD can review from the Microsoft Entra admin portal.
Certificate pin expiry	Alert 60 days and 30 days before the pinned certificate expires, triggering the certificate rotation procedure.

13.3 CI/CD Pipeline

The application uses a GitHub Actions pipeline (or equivalent) with the following stages:

19. Build — compile the Flutter application for Android (release mode).
20. Unit test — run the full Flutter test suite. Build fails if any test fails.
21. Integration test — run integration tests against the staging API environment.

22. Static analysis — Flutter Analyze and a dependency vulnerability scan (e.g. OWASP Dependency Check).
23. Sign — sign the APK with the JCF release keystore. Keystore is stored in the CI secrets vault, not in source control.
24. Distribute (staging) — upload to the internal test track (Google Play Internal Testing or direct MDM distribution).
25. Distribute (production) — promoted manually by ICTD after successful pilot sign-off.

13.4 API Versioning Strategy

The API follows URI versioning (/api/v1/, /api/v2/). The versioning policy is:

- Non-breaking changes (adding optional fields, new endpoints) may be made to an existing version without a version bump.
- Breaking changes (removing fields, changing authentication requirements) require a new version prefix.
- Old versions are supported for a minimum of 90 days after the new version is published. The sunset date is communicated via the Deprecation: and Sunset: response headers.
- The mobile app always targets the latest stable API version. Older app versions that target a deprecated API receive an X-API-Deprecated: true response header, and the app displays an in-app upgrade prompt.

14. Legal and Licensing

This section is new in Version 2.0.

14.1 Third-Party Library Licences

The application is built with Flutter 3 and uses the following principal third-party packages. A full licence inventory is provided in Appendix D.

Package / Library	Licence
Flutter SDK (Google)	BSD 3-Clause
flutter_pdfview (or syncfusion_flutter_pdfviewer)	Apache 2.0 / Syncfusion Community Licence
MSAL for Android (Microsoft)	MIT
OkHttp (Square)	Apache 2.0
Room (Android Jetpack, Google)	Apache 2.0
EncryptedSharedPreferences (AndroidX Security)	Apache 2.0
Firebase Crashlytics (Google)	Apache 2.0 / Google Terms of Service
flutter_tts (Text-to-Speech)	MIT

No package requires the application to be open-sourced or restricts commercial/governmental use. All licences are compatible with deployment in a law-enforcement context. Licence texts will be accessible in the application's About screen.

14.2 Privacy Notice and Data Minimisation

The application is deployed on officers' personal devices. The following personal data is processed:

Data item	Purpose	Retention
Officer email address (@jcf.gov.jm)	Authentication; audit log identity	Duration of active session. Cleared on logout.
Azure AD access token	API authorisation	Up to 1 hour. Cleared on logout.
Azure AD refresh token	Silent token renewal	Up to 90 days. Cleared on logout.
Document access events (audit log)	Security monitoring; compliance	2 years. Stored server-side only. Not on device.
Crash reports (Crashlytics)	Bug detection and resolution	90 days. No PII in crash payloads.
Client IP address	Audit logging; anomaly detection	2 years. Stored server-side only.

The application does not collect: GPS location, device contacts, call logs, SMS messages, biometrics, or any data beyond what is listed above. Officers are informed of this data processing via the in-app privacy notice displayed at first launch, consistent with the JCF's obligations under the Data Protection Act 2020 (Jamaica).

14.3 End User Licence Agreement (EULA)

Officers accept a brief EULA at first launch. Key provisions include:

- The application is provided by the Jamaica Constabulary Force for official use only.
- Document content accessed via the application is classified and must not be reproduced, photographed, or shared outside of authorised JCF channels.
- The officer consents to audit logging of their document access as described in Section 6.5.
- The Force reserves the right to remotely wipe application data from the device.
- Violation of these terms may constitute a disciplinary or criminal offence.

15. Proposed Rollout Plan

Phase	Scope	Duration
1. Pilot	ICTD staff plus one nominated division. JCF webmail sign-in enabled. AD group access control active from day one. Feedback collected via structured survey.	4 weeks
2. Hardening	Add screenshot blocking, watermarking, audit log review dashboard, and load testing based on pilot findings. iOS port evaluated if officer demand exists.	3 weeks
3. Force-wide release	Distribute via MDM or internal application catalogue. Training note circulated. EULA presented on first launch.	2 weeks
4. Ongoing operations	ICTD assumes ownership of content updates and AD group management. Quarterly application updates. Annual security review of access control matrix.	Continuous

16. Request to JCF

The author respectfully requests the following from the Jamaica Constabulary Force:

26. Approval in principle. Authorisation to proceed to a controlled pilot.
27. ICTD sponsorship. Registration of the application within the JCF Microsoft 365 tenant to enable webmail-based authentication and creation of the JCF-DocHub-* AD security groups described in Section 5.
28. A nominated Content Owner. Identification of the officer or unit authorised to add, update, and retire policy documents and to manage AD group assignments.
29. A pilot division. Identification of one division willing to use the application for four weeks and provide structured feedback.
30. Intune MAM enrolment. Authorisation to register the application with the JCF Intune tenant to enable selective app wipe capability.
31. Confirmation of SharePoint as document store. Identification of the SharePoint site or document library that will serve as the ICTD-controlled document source.

No financial outlay is requested for the pilot phase. The application has been developed at the author's own expense, and the supporting infrastructure is leveraged from systems the Force already operates.

17. Closing Statement

Every serving officer carries a smartphone. Every serving officer is expected to know Force policy. The JCF Document Hub closes the gap between those two facts — securely, with access control, with a full audit trail, and with a clear technical foundation that ICTD can own, extend, and monitor.

Version 2.0 of this proposal addresses each technical and governance concern raised during ICTD review. The application is ready to pilot.

Appendix A — Document Categories

The application currently organises 346 documents across the following 19 categories.

Category	Description
Force Orders	Documents from JCF Headquarters governing operational conduct.
JCF Policies	General Jamaica Constabulary Force policies and SOPs.
NPCJ	National Police College of Jamaica training and curriculum material.
TMMD	Traffic and Mobile Maintenance Division procedures.
PMMD	Public Morality and Major Decency Division procedures.
CIB	Criminal Investigation Branch procedures and guidelines.
PECC	Police Emergency Communications Centre procedures.
PRDB	Police Records and Data Branch procedures.
PMAS	Property Management and Administration Services procedures.
FLPD	Firearm Licensing and Permits Department procedures.
FIPT	Firearm Investigation and Prosecution Team procedures.
DWTT	Detective and Witness Testimony Training material.
FIBUA	Fighting in Built-Up Areas tactical material. Restricted to tactical units.
PPMU	Policy and Procedure Management Unit material.
ICTD	Information and Communications Technology Division procedures.
ICT	General Information and Communications Technology guidelines.
SOPs	Cross-divisional Standard Operating Procedures.
SIMU	Strategic Intelligence Management Unit procedures. Restricted — ACP authorisation.
General	Statutes, Acts, and supporting reference material.

Appendix B — AD Group to Document Access Matrix

This appendix is new in Version 2.0. It shows which AD security groups are required to access each document category. Officers are provisioned into groups by ICTD based on their assignment.

Read as: an officer must be a member of the listed group to see documents in that category. The JCF-DocHub-Admin group grants access to all categories.

Document Category	Required AD Group(s)
General	JCF-DocHub-General (all officers automatically assigned on account creation)
JCF Policies / Force Orders	JCF-DocHub-JCF (all officers automatically assigned on account creation)
SOPs	JCF-DocHub-SOPs (all officers automatically assigned on account creation)
ICT	JCF-DocHub-ICT (all officers automatically assigned on account creation)
NPCJ	JCF-DocHub-NPCJ (assigned by NPCJ registrar on enrolment)
TMMD	JCF-DocHub-TMMD (assigned by TMMD OC on posting)
PMMD	JCF-DocHub-PMMD (assigned by PMMD OC on posting)
CIB	JCF-DocHub-CIB (assigned by CIB commanding officer on posting)
PECC	JCF-DocHub-PECC (assigned by PECC supervisor on posting)
PRDB	JCF-DocHub-PRDB (assigned by PRDB officer in charge on posting)
PMAS	JCF-DocHub-PMAS (assigned by PMAS officer in charge on posting)
FLPD	JCF-DocHub-FLPD (assigned by FLPD officer in charge on posting)
FIPT	JCF-DocHub-FIPT (assigned by FIPT team leader on posting)
DWTT	JCF-DocHub-DWTT (assigned by CIB commanding officer — CIB officers and training staff only)
FIBUA	JCF-DocHub-FIBUA (requires written authorisation from SSP Operations)
PPMU	JCF-DocHub-PPMU (assigned by PPMU head)
ICTD	JCF-DocHub-ICTD (assigned by ICTD director — ICTD staff only)
SIMU	JCF-DocHub-SIMU (requires written authorisation from ACP Intelligence)
All categories	JCF-DocHub-Admin (ICTD administrators and nominated Content

Document Category	Required AD Group(s)
	Owner only)

Appendix C — API Endpoint Reference

This appendix is new in Version 2.0. It provides a concise summary of all API endpoints for ICTD technical review. Full request/response schemas will be published as an OpenAPI 3.0 specification prior to pilot commencement.

Method + Path	Auth required	Description
GET /api/v1/categories	Bearer token	List authorised categories. Paginated.
GET /api/v1/categories/{id}/documents	Bearer token	List documents in category. Paginated. Sortable.
GET /api/v1/documents/search?q=	Bearer token	Full-text search. Results filtered to authorised categories.
GET /api/v1/documents/{id}/metadata	Bearer token	Document metadata: title, size, date, hash, version.
GET /api/v1/documents/{id}/content	Bearer token	PDF stream. Supports HTTP Range.
GET /api/v1/documents/{id}/thumbnail	Bearer token	First-page JPEG thumbnail (256x256 px, cached 24 h).
GET /api/v1/sync/manifest	Bearer token	Document hash manifest for offline cache invalidation.
POST /api/v1/audit/access	Bearer token	Record a document access event.

All endpoints return JSON (application/json) unless otherwise noted. The /content endpoint returns application/pdf. All endpoints return the following error envelope on failure:

```
{ "error": "error_code", "message": "Human-readable description", "request_id": "UUID" }
```

HTTP Status	Meaning
200 OK	Request succeeded.
206 Partial Content	Byte-range request succeeded (PDF streaming).
400 Bad Request	Malformed request (invalid document ID, missing parameters).
401 Unauthorised	No token or invalid/expired token.
403 Forbidden	Valid token but insufficient group membership.
404 Not Found	Document or category does not exist.
429 Too Many Requests	Rate limit exceeded. Includes Retry-After header.
500 Internal Server Error	Unexpected server error. Logged and alerted automatically.
503 Service Unavailable	Server overloaded. Includes Retry-After header.

Appendix D — Third-Party Library Licences

This appendix is new in Version 2.0. All third-party packages used in the application are listed below with their licence type and any relevant conditions.

Package	Licence	Conditions
Flutter SDK (Google)	BSD 3-Clause	Attribution required. No copyleft obligation.
Dart SDK (Google)	BSD 3-Clause	Attribution required. No copyleft obligation.
msal_mobile (Microsoft Authentication Library)	MIT	No restrictions on governmental use.
syncfusion_flutter_pdfviewer	Syncfusion Community Licence	Free for government/non-profit use with fewer than 5 developers. ICTD to confirm eligibility. Commercial licence available at low cost if required.
flutter_tts (Text-to-Speech)	MIT	No restrictions.
http (Dart HTTP client)	BSD 3-Clause	No restrictions.
flutter_secure_storage (Keystore/Keychain wrapper)	BSD 3-Clause	No restrictions.
sqflite (local encrypted database)	MIT	No restrictions.
sqflite_cipher (AES-256 encryption for local DB)	MIT	No restrictions.
firebase_crashlytics (Google)	Apache 2.0 + Google ToS	Crash data processed by Google. No PII transmitted. See Section 14.2.
provider / riverpod (state management)	MIT	No restrictions.
intl (internationalisation)	BSD 3-Clause	No restrictions.

Licence texts are bundled in the application and accessible via Settings > About > Open-source licences.

Appendix E — Privacy Notice and Data Minimisation Statement

This appendix is new in Version 2.0. It is intended to serve as the in-app privacy notice shown to officers at first launch.

JCF Document Hub — Privacy Notice

Version 2.0 | Effective: 13 May 2026 | Issued by: Jamaica Constabulary Force (ICTD)

What this application does

The JCF Document Hub provides serving officers with secure access to Force policies, Force Orders, and Standard Operating Procedures on their personal smartphones. It is operated by the Jamaica Constabulary Force, Information and Communications Technology Division.

Data we collect and why

Data	Why we collect it	Where it is stored
Your name @jcf.gov.jm email address	To identify you and control your access to authorised documents.	On your device (during session only). In server-side audit logs for 2 years.
Authentication tokens (issued by Microsoft 365)	To verify your identity on each request to our servers.	On your device in Android Keystore-protected storage. Deleted on logout.
Records of which documents you open or download	For security monitoring and compliance with JCF data governance requirements.	On our servers only. Not on your device. Retained for 2 years.
Crash reports (if a crash occurs)	To identify and fix application defects.	On Google Firebase servers for 90 days. No personal information is included.

Data we do not collect

We do not collect your GPS location, contacts, call logs, SMS messages, photos, biometric data, browsing history, or any other data from your device.

Your rights

Under the Data Protection Act 2020 (Jamaica) you have the right to access the personal data we hold about you and to request correction of inaccurate data. To exercise these rights, contact the JCF Data Protection Officer at dpo@jcf.gov.jm.

Remote wipe

By installing and using this application, you acknowledge that the Jamaica Constabulary Force may remotely delete application data (documents and tokens) from your device in the event of loss, theft, or separation from the Force. This action does not affect your personal data outside the application.

Contact

ICTD Help Desk: ictd@jcf.gov.jm | JCF Data Protection Officer: dpo@jcf.gov.jm

Appendix F — Glossary

Term	Definition
JCF	Jamaica Constabulary Force.
ICTD	Information and Communications Technology Division (of the JCF).
SOP	Standard Operating Procedure.
Force Order	An official instruction issued by JCF Headquarters governing operational conduct.
OAuth 2.0 / OIDC	Industry-standard protocols for delegated authorisation and federated identity.
PKCE	Proof Key for Code Exchange — an extension to OAuth 2.0 that prevents authorisation code interception attacks.
MSAL	Microsoft Authentication Library — the official client library for signing in to Microsoft 365.
Conditional Access	A Microsoft 365 capability that enforces sign-in policies (MFA, device compliance, location restrictions).
Android Keystore	A hardware-backed secure enclave on Android devices for storing cryptographic keys. Keys cannot be extracted even by privileged applications.
MDM	Mobile Device Management — central management and deployment of applications to organisational devices.
MAM	Mobile Application Management — a subset of MDM focused on managing application-level data without controlling the whole device.
Intune	Microsoft's MDM and endpoint compliance product.
TLS	Transport Layer Security — the cryptographic protocol that secures data in transit over HTTPS.
Certificate pinning	A technique where the mobile app is programmed to trust only a specific server certificate, preventing MITM attacks even on devices with rogue root CAs installed.
TTL	Time to Live — the maximum age of a cached item before it must be refreshed from the source.
SHA-256	A cryptographic hash function used to verify document integrity. If a document's hash changes, the content has changed.
AES-256-GCM	Advanced Encryption Standard with 256-bit key in Galois/Counter Mode — the encryption algorithm used for locally cached PDF files.
REST	Representational State Transfer — an architectural style for designing networked APIs using standard HTTP methods.
JWKS	JSON Web Key Set — the public keys published by Microsoft 365 that the backend uses to verify token signatures.

Term	Definition
EULA	End User Licence Agreement — the legal agreement an officer accepts when first using the application.
CDN	Content Delivery Network — a geographically distributed network of servers that caches and serves static content (PDFs, thumbnails) close to the user.
CI/CD	Continuous Integration / Continuous Deployment — automated pipelines that build, test, and release software.

Appendix G — Application Screenshots

The following screenshots were captured from the production build of the JCF Document Hub application running on an Android device. They illustrate the home screen, category navigation, document listings, and the integrated PDF viewer.

Note: Screenshots show Version 1.0 of the application. Version 2.0 will add the following visible UI changes:

- Offline / Cached / Expiring / Updated badges on document tiles (Section 10.4).
- Offline banner in the application header when no connection is available.
- Two-pane layout on tablets (Section 11.4).
- Privacy notice and EULA displayed at first launch (Section 14.3).

[Screenshot figures C.1 through C.3 from Version 1.0 are retained here. Updated screenshots will be provided prior to pilot commencement.]